

# Human Digital Twin for Behavior Simulation Using Artificial Intelligence

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**Abstract**—Human behavior simulation remains a challenging problem in artificial intelligence due to the complex interaction between emotional states and contextual conditions. Existing systems primarily focus on isolated tasks such as emotion recognition or decision prediction, limiting their ability to simulate realistic human decision-making behavior. This paper presents a Human Digital Twin (HDT) framework integrating emotion classification, contextual modeling, and behavioral prediction within a unified architecture. The proposed system utilizes TF-IDF feature extraction and Logistic Regression for emotion classification using the GoEmotions dataset. To address the absence of publicly available datasets combining emotion, context, and behavioral outcomes, a structured synthetic behavioral dataset is generated for training the decision prediction model. Emotional and contextual features are integrated to simulate decision tendencies while maintaining interpretability through behavioral explanation and interpretation modules. Experimental evaluation demonstrates strong behavioral consistency within controlled environments while highlighting the complexity of real-world emotion classification. The proposed framework contributes toward interpretable and human-centric behavioral simulation systems.

**Index Terms**—Human Digital Twin, Artificial Intelligence, Emotion Classification, Behavioral Simulation, Machine Learning, Interpretable AI

## I. INTRODUCTION

Human decision-making behavior is influenced by emotional states, situational conditions, environmental pressure, and contextual reasoning. Modeling such complex behavioral interactions within artificial intelligence systems remains a significant challenge due to the dynamic nature of human cognition and emotional variability.

Existing approaches generally focus on isolated tasks such as emotion recognition, sentiment analysis, or contextual decision prediction. While these methods achieve acceptable performance in specific domains, they fail to capture the combined influence of emotion and contextual reasoning on behavioral outcomes.

The concept of Digital Twin has been extensively explored for modeling physical systems within virtual environments for monitoring, prediction, and optimization [1]. Extending this concept toward human-centric systems enables the development of Human Digital Twin (HDT) frameworks capable of simulating cognitive and behavioral responses.

This paper proposes a Human Digital Twin framework integrating emotion detection, contextual modeling, behavioral prediction, and interpretability within a unified architecture. The proposed system combines TF-IDF feature extraction,

Logistic Regression emotion classification, synthetic behavioral modeling, and contextual feature integration to simulate realistic human decision-making behavior.

The major contributions of this work are summarized as follows:

- Development of a unified Human Digital Twin framework integrating emotional and contextual reasoning.
- Construction of a synthetic behavioral dataset for decision modeling.
- Integration of behavioral interpretation mechanisms for transparent prediction analysis.
- Real-time behavioral simulation using an interactive AI-based interface.

## II. RELATED WORK

The concept of Digital Twin was introduced to replicate physical systems within virtual environments for predictive monitoring and optimization [1], [2]. Recent advancements have extended this concept toward intelligent systems and human-centric modeling.

Emotion recognition using machine learning and natural language processing has gained significant attention in recent years. The GoEmotions dataset introduced by Demszky et al. [3] enables fine-grained emotion classification for textual analysis.

Human-centered artificial intelligence emphasizes reliability, interpretability, and transparency in AI systems [4]. However, most existing approaches independently address emotion classification and decision prediction without integrating contextual behavioral reasoning.

The proposed Human Digital Twin framework addresses this limitation by combining emotion detection, contextual integration, behavioral prediction, and interpretation within a single architecture.

## III. PROPOSED METHODOLOGY

The proposed Human Digital Twin framework consists of emotion classification, contextual integration, behavioral prediction, and behavioral interpretation modules.

### A. Emotion Classification

User textual input is preprocessed and transformed using TF-IDF vectorization.

$$TFIDF(t, d) = TF(t, d) \times IDF(t) \quad (1)$$

where:

$$IDF(t) = \log \left( \frac{N}{df(t)} \right) \quad (2)$$

A Logistic Regression classifier is used for emotion prediction.

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^T x + b)}} \quad (3)$$

### B. Contextual Modeling

Contextual parameters including scenario type and difficulty level are integrated with emotional states to simulate realistic decision-making conditions.

### C. Behavioral Prediction

A structured synthetic dataset is generated to model relationships between emotional states, contextual conditions, and behavioral outcomes.

The integrated feature representation is expressed as:

$$F = E \cup C \quad (4)$$

where:

- $E$  represents emotional features
- $C$  represents contextual features

The behavioral prediction model estimates decision tendencies using the integrated feature set.

## IV. SYSTEM ARCHITECTURE

Fig. 1 illustrates the architecture of the proposed Human Digital Twin framework.

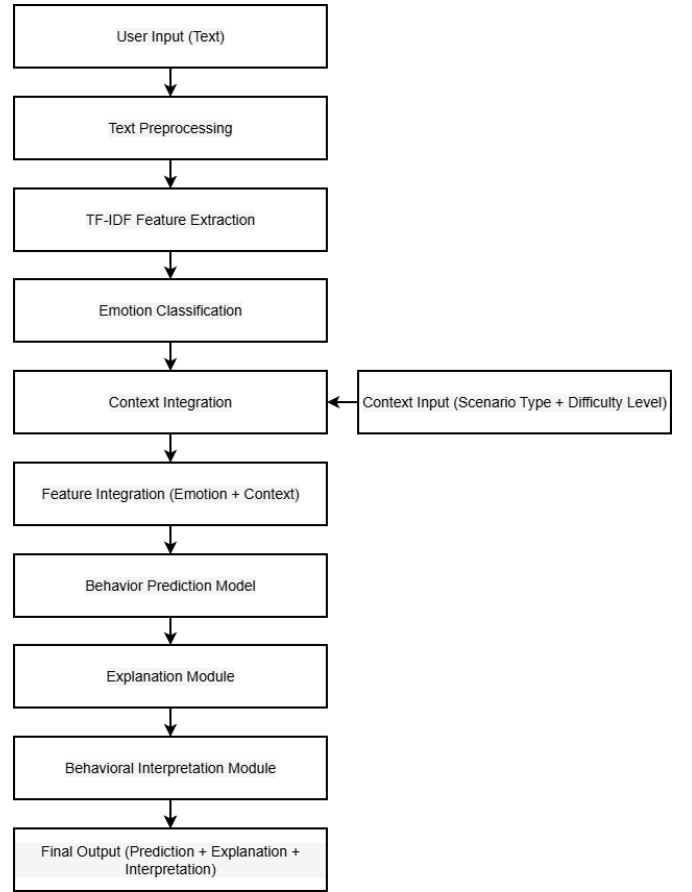


Fig. 1. Architecture of the Proposed Human Digital Twin Framework

The workflow follows:

User Input → Emotion Detection → Context Integration → Behavioral Prediction → Explanation → Interpretation

## V. EXPERIMENTAL RESULTS

### A. Synthetic Dataset Evaluation

The behavioral prediction model achieved an accuracy of 93.09% on the synthetic dataset, demonstrating strong consistency in structured behavioral prediction under controlled contextual conditions.

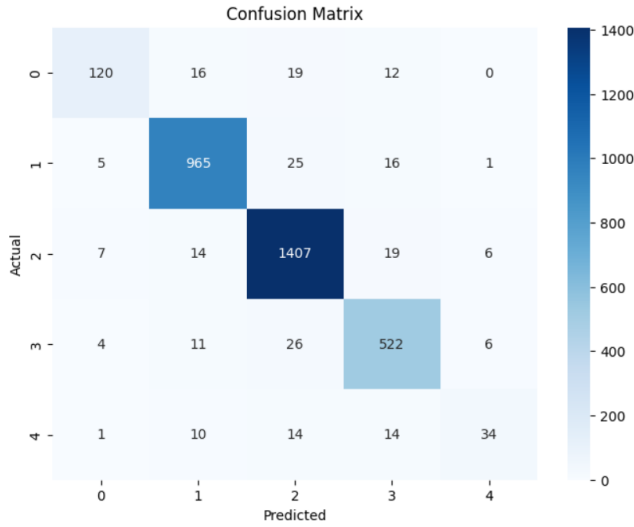


Fig. 2. Confusion Matrix for Synthetic Behavioral Dataset

The confusion matrix demonstrates strong diagonal dominance, indicating stable classification performance.

### B. GoEmotions Evaluation

The GoEmotions-based emotion classification model achieved approximately 3.31% accuracy due to overlapping emotional classes, dataset imbalance, and ambiguity in natural language expressions.

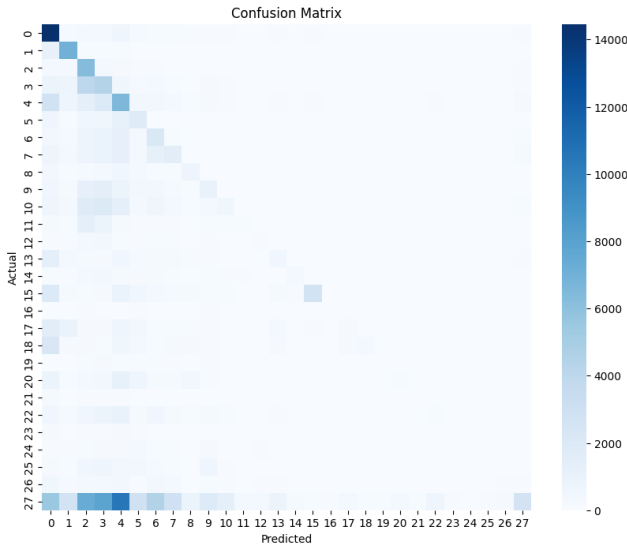


Fig. 3. Confusion Matrix for GoEmotions Dataset

### C. Comparative Analysis

TABLE I  
COMPARATIVE PERFORMANCE ANALYSIS

| Dataset            | Accuracy | Characteristics           |
|--------------------|----------|---------------------------|
| Synthetic Dataset  | 93.09%   | Structured and Controlled |
| GoEmotions Dataset | 3.31%    | Real-world and Complex    |

## VI. REAL-TIME BEHAVIORAL SIMULATION

The proposed framework supports real-time behavioral simulation using a Streamlit-based interactive interface. The system accepts user-defined situations, contextual parameters, and difficulty levels to simulate behavioral outcomes based on emotional and contextual reasoning.

### Human Digital Twin for Behavior Simulation Using AI

HDT System Interface

#### Enter Situation Details

Describe your situation:

I have a new job opportunity but I feel nervous and unsure

Select Scenario:

career

Select Difficulty Level:

high

Fig. 4. Real-Time Human Digital Twin System Interface

The interface enables users to provide contextual input scenarios for behavioral simulation. Emotional states and contextual parameters are processed to generate behavioral predictions and emotional interpretations.

#### Emotion to Behavior Mapping

Detected Emotion: nervousness Mapped Behavioral State: anxious Predicted Decision Tendency: avoidant

#### Explanation

The dominant detected emotion is 'nervousness', which maps to the behavioral state 'anxious'. For the selected scenario 'career' with difficulty level 'high', the predicted decision tendency is 'avoidant'.

#### Behavioral Interpretation

Feeling anxious in challenging situations can increase the tendency to avoid decisions. Higher difficulty levels can increase emotional pressure and affect decision patterns. Using structured planning and breaking the problem into smaller steps may help reduce avoidance.

Fig. 5. Behavioral Explanation and Interpretation Output

The generated output includes:

- Detected emotional state
- Behavioral state mapping
- Predicted decision tendency
- Context-aware explanation
- Behavioral interpretation

## VII. DISCUSSION

The proposed Human Digital Twin framework demonstrates that integrating emotional intelligence with contextual reasoning improves behavioral simulation capability compared to isolated prediction approaches.

The synthetic behavioral dataset provides controlled feature relationships, enabling stable behavioral prediction performance. In contrast, the GoEmotions dataset introduces significant complexity due to overlapping emotional expressions, class imbalance, and ambiguity in natural language.

Although emotion classification accuracy remains comparatively lower, the framework maintains meaningful behavioral prediction through contextual integration and emotional mapping.

The behavioral interpretation module further improves transparency by connecting emotional states, contextual conditions, and predicted behavioral outcomes.

The experimental results validate the feasibility of combining natural language processing, contextual reasoning, and behavioral simulation within a Human Digital Twin framework.

### VIII. LIMITATIONS AND FUTURE WORK

The proposed framework relies on synthetic behavioral data because publicly available datasets integrating emotion, context, and behavioral outcomes remain limited.

Additionally, the emotion classification component is affected by class imbalance and overlapping emotional representations within the GoEmotions dataset.

Future work will focus on:

- Real-world behavioral dataset integration
- Adaptive and personalized Human Digital Twins
- Multi-turn contextual reasoning
- Transformer-based emotion modeling
- Advanced behavioral interpretation mechanisms

### IX. CONCLUSION

This paper presented a Human Digital Twin framework for behavior simulation using emotion detection, contextual modeling, and machine learning-based behavioral prediction.

The proposed system demonstrates the feasibility of integrating emotional intelligence and contextual reasoning to simulate human decision-making behavior within an interpretable AI framework.

The framework successfully combines NLP-based emotion detection, contextual feature integration, and behavioral prediction to generate realistic and explainable behavioral responses.

The proposed Human Digital Twin framework establishes a strong foundation for future research in adaptive, interpretable, and human-centric artificial intelligence systems.

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